CS 441 - Final Project

Group Member Names	NetIDs
Sebastian Chamorro	sc128
Arieb Rana	ariebmr2
Janine Zhu	janinez2
Maya Ashok	mayaa4

1. Problem Description

What does the application do? What problem does it address? (2-4 sentences)

Our application would categorize different events in sports into five main, recognizable labels. These events are specifically focused on specific actions, such as passing (in the respective sport), which would occur in a match/event. This aims to solve the problem of being unfamiliar with a sport. By providing an interface that would match an unfamiliar event to a recognizable label, we aim to make sports lingo more accessible.

Who is the intended user? What is the use case? When would it be used? What is the benefit? (2-4 sentences)

The intended user, as stated above, would be people wishing to enjoy a new sport to either watch or play. There are many different rules, actions, and plays that vary between seemingly similar sports, so we aim to breach that gap. The benefit would be in simplifying complex terminology for new sports fans, and it could be used whenever watching a game where commentary could be fed into our application. There the user would be able to understand under what category that specific action falls into.

2. ML Approach

How did you collect/curate the training and test data? How do you evaluate your approach (e.g. what metrics)?
Make clear the level of effort, innovation, creativity involved, including research and implementation.

The first step in our collection process was the selection of the sports that would ultimately make up our training/test sets. As described in both the final video, as well as portrayed by the various links towards our project resources, we ultimately ended up deciding on the following subdivision: soccer, ufc, tennis, and F1. This was an important part of the process, as we are the “source of input” for the categories. With choosing sports that each one of us was most familiar with, we were more confident that the chosen categories would be the most “in-line” with seasoned fans of each sport. In doing so, any user of the application would be better suited to receive previous knowledge that years of following each sport would provide.

Following this, we aimed to make sure that the events chosen had a mix of both familiar as well as unfamiliar choices, such that the application of this project would be the most helpful at “introducing” a sport to someone. To accomplish this, each of us sought to first find various commentary sources describing games minute by minute. This is a very lengthy, but informative, source of information, that required exterior work in order to scrape solely the events/actions performed by players. In addition, some consideration was put into place in order to include multiple “variations” of common events. For example, the word “passing” in soccer seems to signify a very ambiguous action. In order to overcome this, we researched various sources, ultimately deciding on utilizing ones that included additional descriptive tokens for these events. For that same event, one could have a “diagonal_pass”, which in soccer signifies an offensive diagonal play toward the opponents half of the field. By adding this type of “data-augmentation”, we aimed to make our variance lower, and so improve (or planned to improve) the performance of our model.

Ultimately we decided to split this total data set into training, validation, and test subsets in the following proportion: 80% training and validation, and 20% test.

In order to later evaluate our approach, we used the following categories:

1) Offensive
2) Defensive
3) Transition
4) Error
5) Neutral

We primarily relied on accuracy results in order to evaluate our model. Basically comparing the results output from the various models that we used and determining if it matches the categories that we imposed in the data recollection section. In addition, we also used a qualitative approach

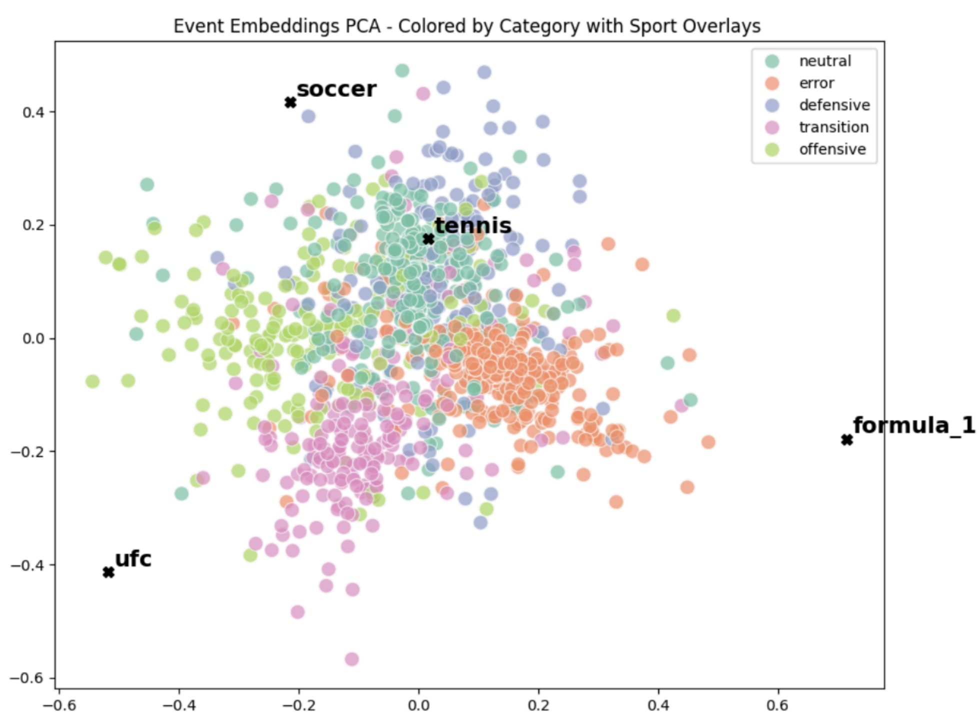
(PCA), that showed the various clusters between sport and category overlays. The combination of these, with a few more tests such as one-shot testing, were used to evaluate our data. Note that we utilized one-shot as well as nearest neighbor for our sports in order to better evaluate beyond simply accuracy the performance of our model for unseen event tokens.

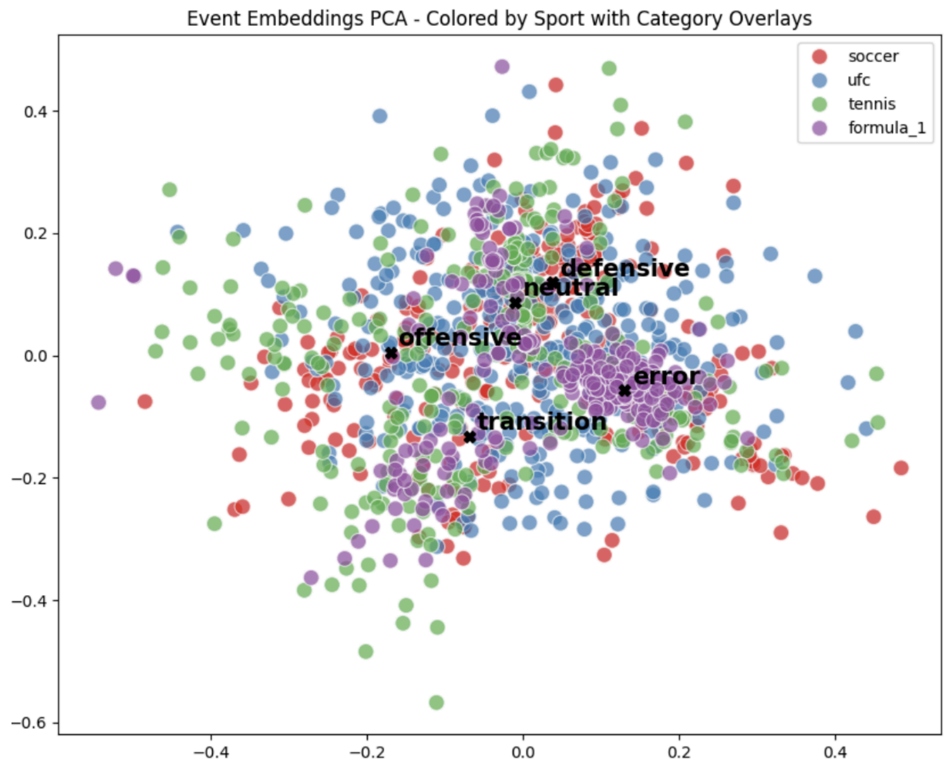
Yet, it is also important to note that the main metric that we used to evaluate our approach was that of the accuracy of the results.

In addition, although a combination of utilizing the models and their accuracy, we implemented a radio interface that would allow for live evaluation of the model. This in mind that it would be a much better interface than running the entire code base for each evaluation.

Machine Learning Model	**Average Test Accuracy +/- Standard Deviation**	**Best Single Run Test Accuracy**
Multi-Layer Perceptron	52.03% +/- 2.99%	57.32%
Logistic Regression	52.03% +/- 2.25%	55.69%
Linear SVM	52.97% +/- 2.07%	55.69%

PCA Plots For MLP Model





Nearest Neighbors (3-NN) Analysis for Sports Data

Nearest Sports for soccer:
1. tennis (Similarity: 0.3243)
2. formula_1 (Similarity: -0.1331)
3. ufc (Similarity: -0.6621)

Nearest Sports for tennis
1. formula_1 (Similarity: 0.4189)
2. soccer (Similarity: 0.3243)
3. ufc (Similarity: -0.8206)

Nearest Sports for ufc:
1. formula_1 (Similarity: -0.5287)
2. soccer (Similarity: -0.6621)
3. tennis (Similarity: -0.8206)

Nearest Sports for formula_1:
1. tennis (Similarity: 0.4189)
2. soccer (Similarity: -0.1331)
3. ufc (Similarity: -0.5287)

Describe the models, hyperparameters, and any preprocessing methods that you tested. (prefer outline format: 1, 1a, 1b, 2, etc)
Make clear the level of effort, innovation, creativity involved, including research and implementation.

1) Pre-Processing:

We first processed the data recollected, whose process is shown above, and was primarily three portions. The first was making the data more “usable”. Since most of our data, as described above, has additional descriptors, we tokenized the events by removing and splitting on underscores.

In addition to this, in order to make a better transition between the tokenized data and our MLP, we decided to include <PAD> and <UNK> tokens. The former is important for the MLP as it receives as inputs batches of data. In order to make sure that all the batches have the same shape, the <PAD> token would “pad” the empty slots. As discussed before, we do utilize zero-shot as a method of evaluating our approach, and the <UNK> token would allow us to make sure that the model would have an “out” in case of unseen/unknown tokens (words).

Also, due the nature of the repeated events, as well as some “unique” events, we decided to use TF-IDF to level this discrepancy. In simple words, this would allow different weights (basically importance) to different events. Thus, tokens that would show up multiple times would be of lower importance than those that would appear multiple times. We did this in order to provide a clearer baseline eventually for our MLP, and be able to generalize it better across the different sports that we chose.

2) MLP (Used Model):

We realized very early on that the data was multi-compositional and most importantly is not linearly separable, thus simple classification and/or regression would not cut it. Thus we used an MLP as our primary model, since it seems to be “small” enough to prevent significant overfitting on our relatively small data set. This had the following hyperparameters:

- Word Embedding Dimension: 16
- Sport Embedding Dimension: 16
- Hidden Layer Size: 64
- Dropout: 0.5
- Optimizer: Adam
- Learning Rate: 0.005
- Weight decay: 5e-3
- Epochs: 100

The MLP would as a goal learn representations of event descriptions and classify them. The general process would be as follows: first the event names are tokenized and mapped to learned word embeddings (basically vector representations of each event). In order to get a fixed-length event representation, we computed IDF which would emphasize informative tokens while downweighting common ones. This would ultimately lead to the events being of the same length. Later, the event and sport embeddings are concatenated and passed through a two-layer

MLP with ReLU activation and dropout regularization. This would then be run through the encoder-decoder to lastly provide us with an accuracy.

3) Logistic Regression/Linear SVM (Baseline Models):

These two models would receive the word embeddings, and perform a baseline performance. As stated previously, it was clear that the data itself would not be fit for a linear classification/regression. Thus, in order to compare our “primary” model with that of other models in order to “proof-of-concept” it we utilized these models and measured its accuracy. As expected, and as to be seen in our application video, these models performed worse than that of our Embedded MLP. We imported these models from the respective scikit-learn libraries and utilized the metrics (sklearn.metrics) in order to evaluate these models.

3. Application Video

Put your link here (e.g. to YouTube or drive). Make sure it’s publicly viewable.

https://youtu.be/INIcfop4ppk

1. Proposal / Sharing

Did you share your project video on CampusWire? (see post #394 for instructions)

Yes

Did you submit the final project form by Nov 20?

Yes

2. Acknowledgments / Attribution

Link to your code (required!!)

https://github.com/sebastian-chamorro/CS441_FINAL_PROJECT

Link to your data (can be a github page or repo)

https://github.com/sebastian-chamorro/CS441_FINAL_PROJECT

External citations or resources

.CSV Data Collection Resources:

- Soccer Event Data:
  - https://github.com/statsbomb/open-data/blob/master/data/events/16079.json
  - https://github.com/statsbomb/open-data/blob/master/data/events/3788741.json
  - https://github.com/statsbomb/open-data/blob/master/data/events/8650.json
  - https://github.com/statsbomb/open-data/blob/master/data/events/8655.json
  - https://github.com/statsbomb/open-data/blob/master/data/events/3788755.json
  - https://github.com/statsbomb/open-data/blob/master/doc/StatsBomb%20Open%20Data%20Specification%20v1.1.pdf
  - https://www.hudl.com/products/wyscout
- F1 Event Data:
  - https://www.autosport.com/f1/live-text/f1-qatar-gp-live-commentary-and-updates-race-1126135/1126135/
  - https://www.fia.com/sites/default/files/fia_2025_formula_1_sporting_regulations_-_issue_1_-_2024-07-31.pdf
  - https://tracinginsights.com/penalty-points/
  - https://www.motorsport.com/f1/news/the-fias-f1-guidelines-for-penalties-and-points-in-full/10736177/
- UFC Event Data:
  - http://ufcstats.com/statistics/events/completed
  - https://www.tapology.com/

- Tennis Event Data:
  - https://www.itftennis.com/en/about-us/organisation/tennis-glossary/
  - https://open.online.uga.edu/tennis/chapter/smashandlob/
  - https://tt.tennis-warehouse.com/index.php
  - https://www.usta.com/en/home/improve/tips-and-instruction/national/tennis-terms-definitions.html#tab=tournaments
  - https://letsgotennis.com/tennis-tips/tennis-terminology/
  - https://en.wikipedia.org/wiki/Glossary_of_tennis_terms

Coding Resources:

- TF-IDF clarification: https://www.geeksforgeeks.org/machine-learning/understanding-tf-idf-term-frequency-inverse-document-frequency/
- MLP: https://scikit-learn.org/stable/modules/neural_networks_supervised.html
- Logistic Regression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html
- Linear SVM: https://scikit-learn.org/stable/modules/svm.html
- PCA: https://www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/

Debugging & general guidance/structure Resources:

- https://chatgpt.com/
- https://gemini.google.com/

Group member contributions

Sebastian - Data Recollection (Soccer), Report
Arieb - Data Recollection (UFC), Report
Maya - Data Recollection (Tennis), Code
Janine - Data Recollection (F1), Video Editing

Did group members contribute roughly equally or unequally? If unequally, explain and specify whether one member went above and beyond, or someone contributed less than agreed or expected.

Equally

Each group member roughly worked equally, with work being done in different portions of the project as outlined in the above section.